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MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
END TERM EXAMINATION-December 2022

Course: B.Tech

Semester: III

Subject Name: Measurement and Instrumentation

Subject Code: EE-214

Time Duration: 3 Hours

Maximum Marks: 60

Subject Coordinator Name (s): Dr. Sushma Gupta & Dr. More Raju

S.No	Question	Marks	CO
1	A) Explain the construction and working principle of LVDT. Explain how the magnitude and direction of displacement of core of an LVDT detected.	5	CO-3
	B) The output of an LVDT is connected to a 5 V voltmeter through an amplifier whose amplification factor is 250. An output of 2 mV appears across the terminal of LVDT when the core moves through a distance of 0.5 mm. Calculate the sensitivity of the LVDT and that of the whole set-up. The milli-voltmeter scale has 100 divisions. The scale can be read to 1/5 of a division. Calculate the resolution of the instrument in mm.	5	CO-3
2	A) Explain Kelvin double bridge and Carey foster slide wire bridge and find out the balanced equation.	5	CO-4
	B) Describe Anderson Bridge and Hay's Bridge with circuit diagram and phasor diagram. Enumerate their advantages and disadvantages.	5	CO-4
3	A) What is seebeck effect and explain the principle of thermocouples?	5	CO-3
	B) Explain strain gauge theory. Derive the expression of gauge factor.	5	CO-3
4	A) Define the following. i) Static Error ii) Precision iii) Sensitivity iv) Accuracy v) linearity	5	CO-1
	B) A moving-coil instrument whose resistance is 25 Ω gives a full scale deflection with a voltage of 25 mV. This instrument is to be used with a series multiplier to extend its range to 10 V. Calculate the error caused by 10 ⁰ C rise	5	CO-1, CO-2

	in temperature. The temperature coefficient of copper is $0.004/^{\circ}\text{C}$ and that of manganin is $0.00015/^{\circ}\text{C}$		
5	A) With help of two wattmeter method, explain how power in three phase circuit can be measured. Support your answer with suitable phasor diagram for star (Wye) connected load. B) The power flowing in a three phase, three wire balanced load system is measured by two wattmeter method. The reading of wattmeter A is 7500 W and of wattmeter B is -1500 W. What is the power factor of the system? C) What is creeping in an energy meter? What is the major cause for it?	5 3 2	CO-2 CO-2 CO-2
6	A) A 1000/5 A, 50 Hz current transformer has a secondary burden comprising a non-inductive impedance of 1.6Ω. The primary winding has one turn. Calculate the flux in the core and ratio error at full load. Neglect leakage reactance and assume the iron loss in the core to be 1.5 W at full load. The magnetizing MMF is 100 A. B) If the voltage $V(t)$ shown in Fig.1 is measured using moving coil and moving iron instruments, find their corresponding readings.	5 5	CO-5 CO-2

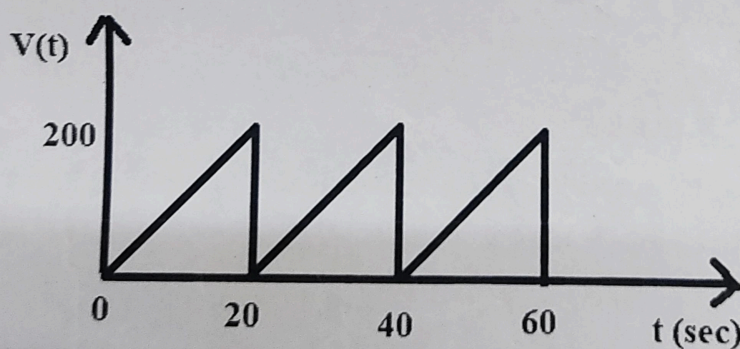


Fig.1